

## Logic Qualifying Exam

### January 2009

Answer six questions and at least one from each section.

#### Section 1

1. Sketch the proof of Godel's Completeness Theorem for Predicate Logic and explain the use of Henkin constants (or Skolem functions).
2. Show that for any elementary chain  $\{\mathcal{A}_i\}_{i \in \omega}$  of structures,  $\mathcal{A}_i$  is an elementary submodel of the union  $\mathcal{A}$ .
3. Show that the theory of dense linear orderings without end points is complete and decidable.

#### Section 2

4. Suppose  $2 \leq \kappa \leq \lambda$  and  $\lambda$  is infinite. Show that  $2^\lambda = \kappa^\lambda$ .
5. State and prove Konig's Lemma. (Hint: trees.)
6. Show that for any notion of forcing  $\mathcal{P}$  and any countable set  $\mathcal{D}$  of  $\mathcal{P}$ -dense sets, there exists a  $\mathcal{D}$ -generic  $\mathcal{P}$ -filter.

#### Section 3

7. Sketch a proof of Godel's incompleteness Theorem for Arithmetic.
8. Define many-one and truth-table reducibility and show that one implies the other but the converse does not hold in general.
9. State the Friedberg-Muchnik Theorem and sketch the proof. Explain how the "priority" argument deals with "injury".